

Program Question_1

```
Implicit none
Integer,Parameter::n=3
Real(8),Parameter::TOL=1.0E-5
Integer,Parameter::M=25
Real(8)::A(n,n), b(n), X0(n), x(n)
Real(8)::norm, sum
Integer::i, j, k

!Initialize the Matrix A

A = reshape([ &
  2.0, -1.0, 1.0, &
  2.0, 2.0, 2.0, &
  -1.0, -1.0, 2.0], [n, n])

!Initialize Vector b
b=[-1.0, 4.0, -5.0]
!Initalize guess
X0=0.0
x=0.0
Write(*,*)"Iter      x1      x2      x3"
Write(*,*)"_____"
```

!step-1

k=1

!step-2

```
Do While(k<=M)
  norm=0.0
  !step_3(compute new x(i)
  Do i=1, n
    sum=0.0
    Do j=1, n
      If(j/=i)Then
        sum=sum+A(i, j)*X0(j)
      End If
    End Do
    x(i)=(b(i)-sum)/A(i, i)
  End Do
  !Step-4(Compute Infinity norm)
  Write(*, ' (I4, 3X, F9.5, 3X, F9.5, 3X, F9.5)' )k, x

  !Print the urrent iteration

  !step-5
  k=k+1
  !step6
  X0=x

End do
Write(*,*)"after 25 Iteration"

Write(*,*)x
```

End Program